Question	Answer	Marks
2(a)(i)	mean/average square speed/velocity	**B1**
2(a)(ii)	$pV = NkT$ **or** $pV = nRT$	**B1**
	$\rho = Nm / V$ **or** $\rho = nN_A m / V$ **and** $k = nR / N$	**B1**
	$E_K = ½ m\langle c^2\rangle$ with algebra to $(3 / 2)kT$	**B1**
2(b)(i)	no (external) work done **or** $\Delta U = q$ **or** $w = 0$	**B1**
	$q = N_A \times (3 / 2)k \times 1.0$	**M1**
	$N_A k = R$ so $q = (3 / 2)R$	**A1**
2(b)(ii)	specific heat capacity $= \{(3 / 2) \times R\} / 0.028$	**C1**
	$= 450\ \text{J kg}^{-1}\ \text{K}^{-1}$	**A1**